

Random Hyperbolic Surface Spectral Rigidity — cycles 13-15

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Random Hyperbolic Surface Spectral Rigidity — cycles 13-15

Abstract

Cycles 13-15 opened and validated the M5-extension-candidates milestone. The campaign moved from reconstructing and probing Kim-Tao’s random-cover rigidity argument toward a concrete extension program centered on the Markov/interpolation bottleneck identified earlier in the proof ledger.

The three-cycle arc was:

1. Cycle 13 ranked extension candidates and selected a primary path: isolate when Markov-type derivative amplification is a true structural obstacle and when it reflects unstable reconstruction from sparse reciprocal- n data.
2. Cycle 14 tested the primary path on fixed labelled templates using the M4 falling-factorial expectation identity. It showed that normalized fixed-template expectations have explicit Taylor expansions at $x=1/n$ and are analytically stable near $x=0$.
3. Cycle 15 varied template size. It showed that growing count profiles can produce large coefficient and derivative growth because falling-factorial zero/pole scales move toward $x=0$ like $1/L$.

The result is not an improvement to the Kim-Tao rigidity exponent. It is a validated toy benchmark principle: fixed conflict-free labelled-template expectations are stable after normalization, while Kim-Tao-like derivative amplification first reappears when the size, support, or constraint profile grows.

Introduction

The long-exposure campaign studies Kim and Tao’s paper 2603.01127, “Eigenvalue rigidity of hyperbolic surfaces in the random cover model.” Earlier cycles established the proof archi-

texture, reconstructed the main rigidity and delocalization arguments, built computational probes, and certified a finite labelled-template expectation identity.

Cycles 13-15 focused on the fifth milestone, M5-extension-candidates: finding a credible research direction that builds on the validated internal evidence without claiming more than the artifacts support.

The central technical object in these cycles is a normalized labelled-template expectation. For a conflict-free labelled template with vertex count V and per-generator constraint counts C_a , the M4 identity gives

$$E_H(n) = (n)_V / \text{Product}_a (n)_{C_a}.$$

Here $(n)_k$ is the falling factorial. The normalized observable is

$$N_H(n) = n^{C-V} E_H(n), \quad C = \sum_a C_a.$$

After substituting $x=1/n$, the observable becomes a finite product or ratio of factors $(1 - jx)$. This makes fixed-template Taylor expansions explicit. Cycles 14-15 used this identity as a controlled model for the interpolation losses previously identified in the Kim-Tao proof ledger.

Approach

The cycles used a staged progression.

Cycle 13 was analytical. It inventoried validated evidence from M2, M3, and M4, ranked seven possible extension directions, and selected one primary target. The output was a ranked candidate matrix, a CSV score table, and a primary candidate statement.

Cycle 14 was symbolic and comparative. It expanded fixed-template normalized expectations through order x^4 , compared them with Cycle 9 polynomial-window fits, and tested whether high-degree fit instability came from the exact expectation or from sparse-grid interpolation.

Cycle 15 was symbolic and computational. It varied the profile size parameter L , expanded growing count-profile families through order x^8 , and measured coefficient growth, derivative growth, and nearest falling-factorial singularity scales.

All three cycles preserved conservative scope. The work is about finite permutation and labelled-template mechanisms that model one bottleneck in the Kim-Tao proof architecture. It does not certify Selberg trace formula inputs, MPvH/Nau/MP23 estimates, or a new rigidity theorem.

Source Inventory and Timeline

Cycle	Source sessions	Main artifacts	Decision
13	Researcher d1abcc72-61a9-4c04-87c4-9d033a1fdb4; worker 97208720-81aa-472a-9a23-6c75eb3a4ab5; auditor 9b4dad9-3ce9-4d3b-9dff-6b5ad699c58c	m5_extension_candidate_ranking.md, m5_primary_candidate_statement.md, m5_candidate_scores.csv, m5_extension_candidate_matrix.png	Validated M5 candidate ranking and selected Markov/interpolation-loss sharpening as primary.
14	Researcher a24e3475-402a-40e6-842f-a64f72049074; worker 17268f9e-adbc-44ae-90f5-0dce6fd335dd; auditor 30702dc0-9920-45f8-8f36-b8e9d6c310c6	derive_labelled_embedding_expansions.wls, compare_expansions_to_cycle9.py, expansion CSVs, m5_expansion_vs_cycle9_fits.png, m5_falling_factorial_expansion_test.md	Validated fixed-template expansion benchmark.
15	Researcher bcd1ac47-e279-449f-a77d-dff1d48cbbd5; worker 66c94dc9-f429-4d34-84f4-d9ef7d8e2f33; auditor bb04c33d-3042-4fad-a335-eaee10bdea9f	probe_growing_template_expansions.wls, plot_growing_template_expansions.py, growing-template CSVs, three figures, m5_growing_template_expansion_growth.md	Validated growing-template amplification benchmark.

No REFERENCES.md file was present in the workspace. The references section therefore lists local source artifacts and session IDs instead of global numbered citations.

Findings

Finding 1: The Best M5 Path Is the Markov/Interpolation Bottleneck

Cycle 13 ranked seven extension candidates:

Rank	Candidate	Decision
1	Markov/interpolation-loss sharpening	advance-primary
2	Exact labelled-template polynomial expansion	advance-technical-lemma
3	Delocalization-side improvement	advance-secondary
4	Schreier spectral-window benchmark	advance-benchmark
5	Tree closed-walk moment certification	defer-supporting

Rank	Candidate	Decision
6	Direct Weil-Petersson transfer	reject-defer
7	Full Selberg trace formalization	reject-defer

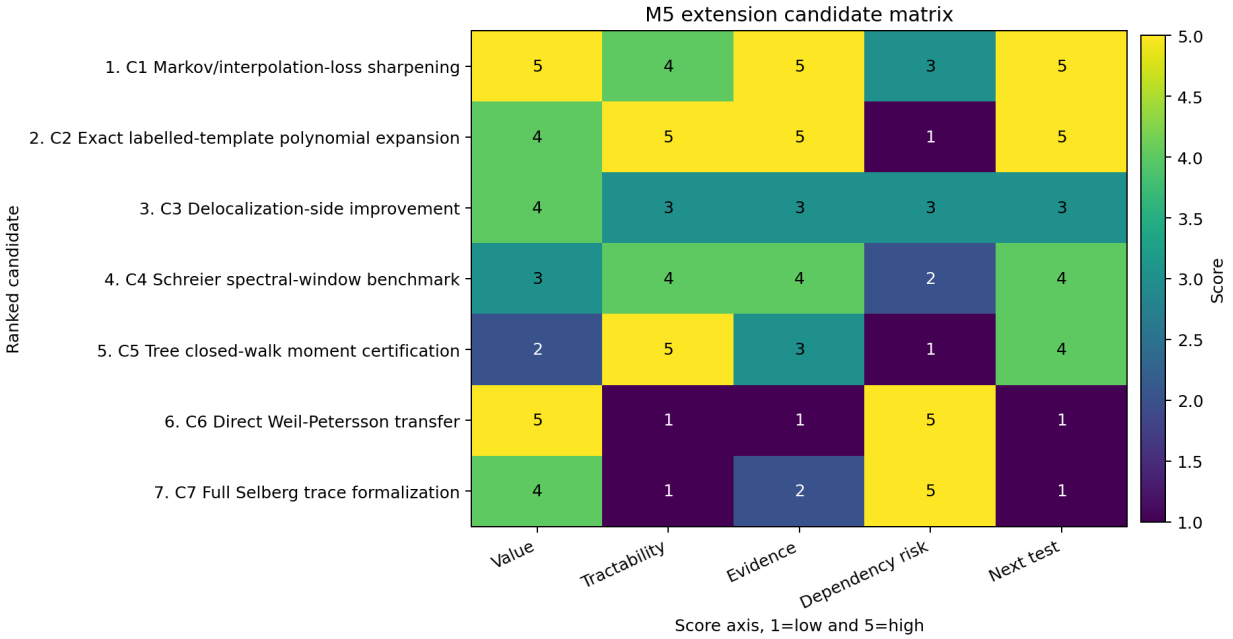


Figure 1: Ranked M5 candidate matrix comparing value, tractability, evidence strength, dependency risk, and next-test clarity.

The decision came from three prior validated facts:

- M2 identified Markov-type amplification in both the trace-side and delocalization-side arguments.
- M3 showed low-degree labelled-template observables were stable while high-degree sparse-grid fits were ill-conditioned.
- M4 certified the exact finite falling-factorial identity underlying one labelled-template class.

The primary candidate was stated conservatively: fixed conflict-free labelled templates should have stable low-degree expansions in $x=1/n$; high-degree sparse-grid instability should not be treated as evidence that the exact expectation is unstable until checked against exact expansions.

Finding 2: Fixed-Template Expectations Are Analytically Stable

Cycle 14 derived exact normalized expansions through x^4 .

Template	Expansion
single_label_cycle	1

Template	Expansion
eight_word_cyclic_toy	1
trace_pair_toy	$1 - x - x^2 - x^3 - x^4 + 0(x^5)$
eight_word_rank2_toy	$1 - 9x + 9x^2 + 39x^3 + 81x^4 + 0(x^5)$
no_edge_control	$1 - x$
single_edge_control	$1 - x$
conflicting_domain	0

For the key rank-two benchmark,

$$N_H(n) = n(n-7)/(n-4)^2 = ((n-6)(n-5)(n-4)) / ((n-1)(n-2)(n-3)).$$

The Cycle 14 audit confirmed that the Wolfram symbolic path and Python comparison path agreed. It also confirmed that the conflict template vanished exactly and that the recovered trace_pair_toy had coefficients [1, -1, -1, -1, -1].

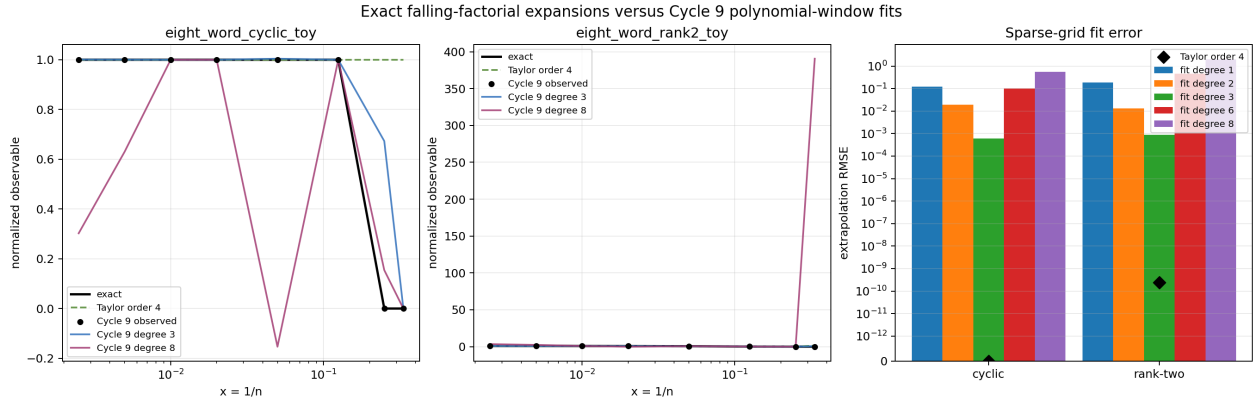


Figure 2: Exact falling-factorial expansions versus Cycle 9 polynomial-window fits for cyclic and rank-two eight-word templates.

The comparison supported the mechanism split. For eight_word_rank2_toy, the exact order-4 Taylor extrapolation RMSE was 2.45e-10, while the Cycle 9 degree-3 extrapolation RMSE was 8.90e-4; degree 6 and degree 8 fits were much less stable. The exact fixed-template expectation is therefore not the source of the high-degree blowup observed in sparse-grid fitting.

Finding 3: Growing Template Size Reintroduces Derivative Amplification

Cycle 15 changed the axis from fixed templates to growing count profiles. The same M4 identity was used, but v and the constraint counts C_a grew with a size parameter L .

Two hand-checkable families behaved benignly:

```
single_label_cycle_profile: N_L(1/x) = 1
single_label_path_profile: N_L(1/x) = 1 - Lx
```

Nontrivial rank-two and rank-four profiles showed large coefficient and derivative growth by $L=40$.

Family	L	L1 coefficient norm through order 8	Max derivative through order 8	Radius proxy
single_label_ cycle_profile	40	1	1	exact cancellation
single_label_ path_profile	40	41	40	1/40
rank2_ balanced_ profile	40	1.222740960544758e20	4.887997076237615e24	1/78
rank2_ deficit_k3	40	3.9254589694304007e19	1.5663067375849087e24	1/76
rank4_ delocalization_ toy_s4	40	6.802812940161747e26	2.7401689056357774e31	1/155

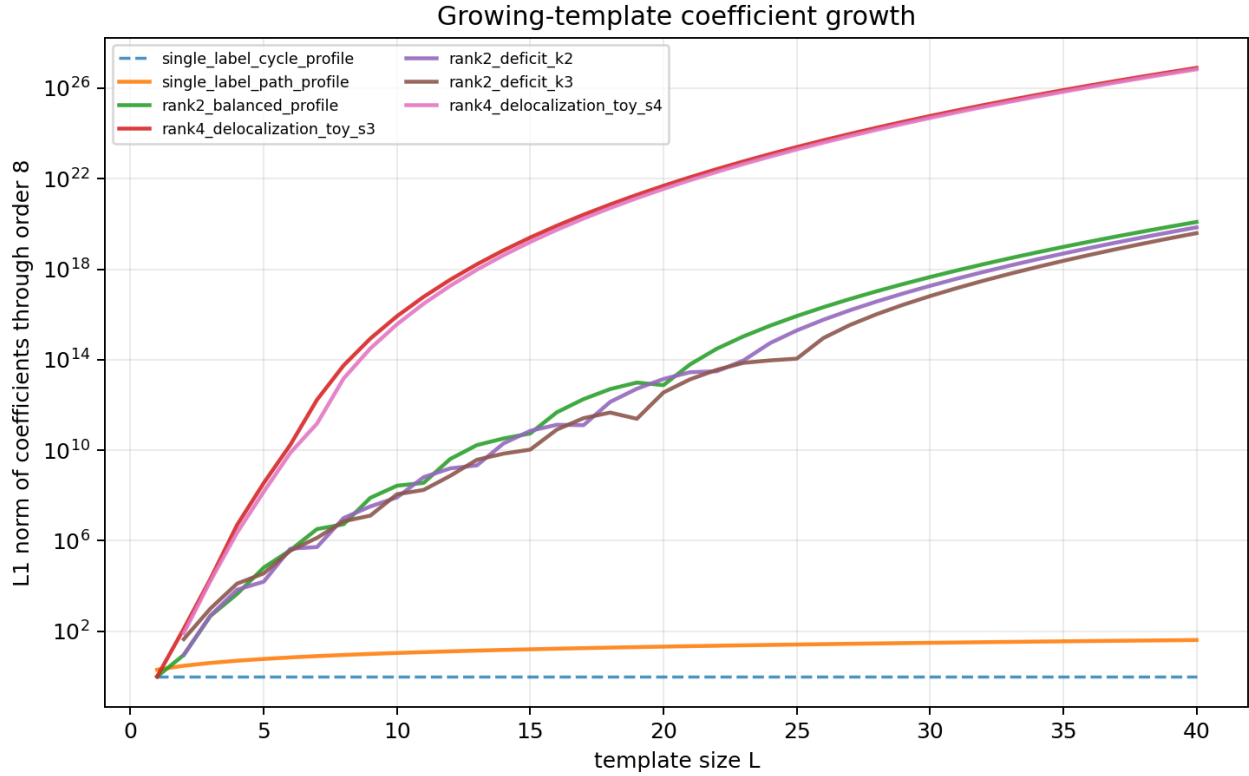


Figure 3: Coefficient norms through order 8 versus template size L for controlled labelled-template profiles.

The mechanism was localized to the falling-factorial product structure. As L grows, the nearest zero or pole scale approaches $x=0$, roughly like $1/L$.

The Cycle 15 audit validated the result and corrected one stale worker note: although the worker reported a local Wolfram license failure, the auditor successfully ran `wolfram-batch-script scripts/probe_growing_template_expansions.wls` and regenerated the CSVs.

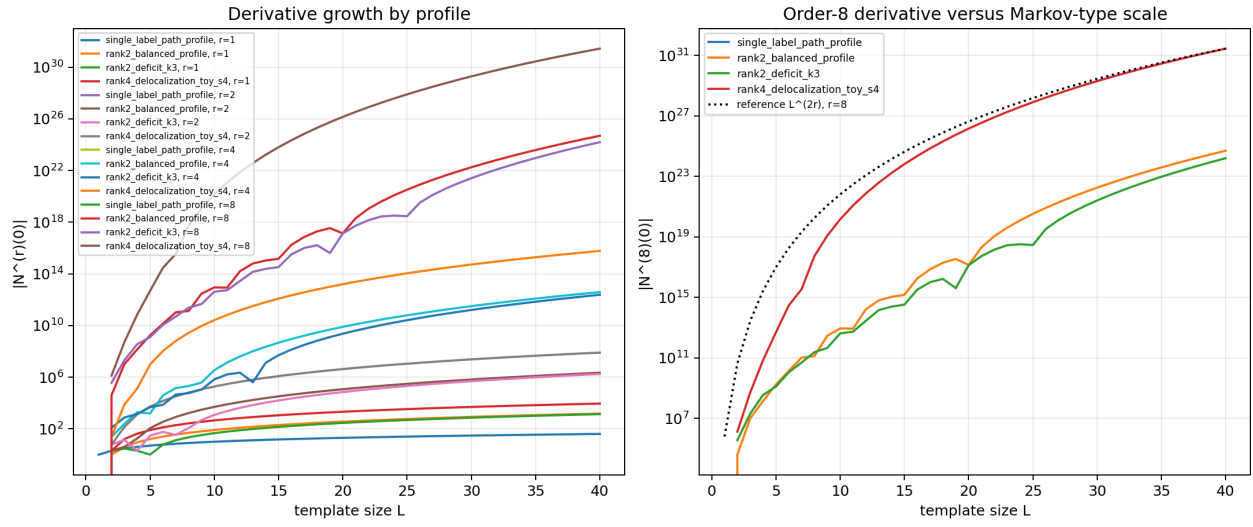


Figure 4: Derivatives at $x=0$ by expansion order for growing labelled-template profiles.

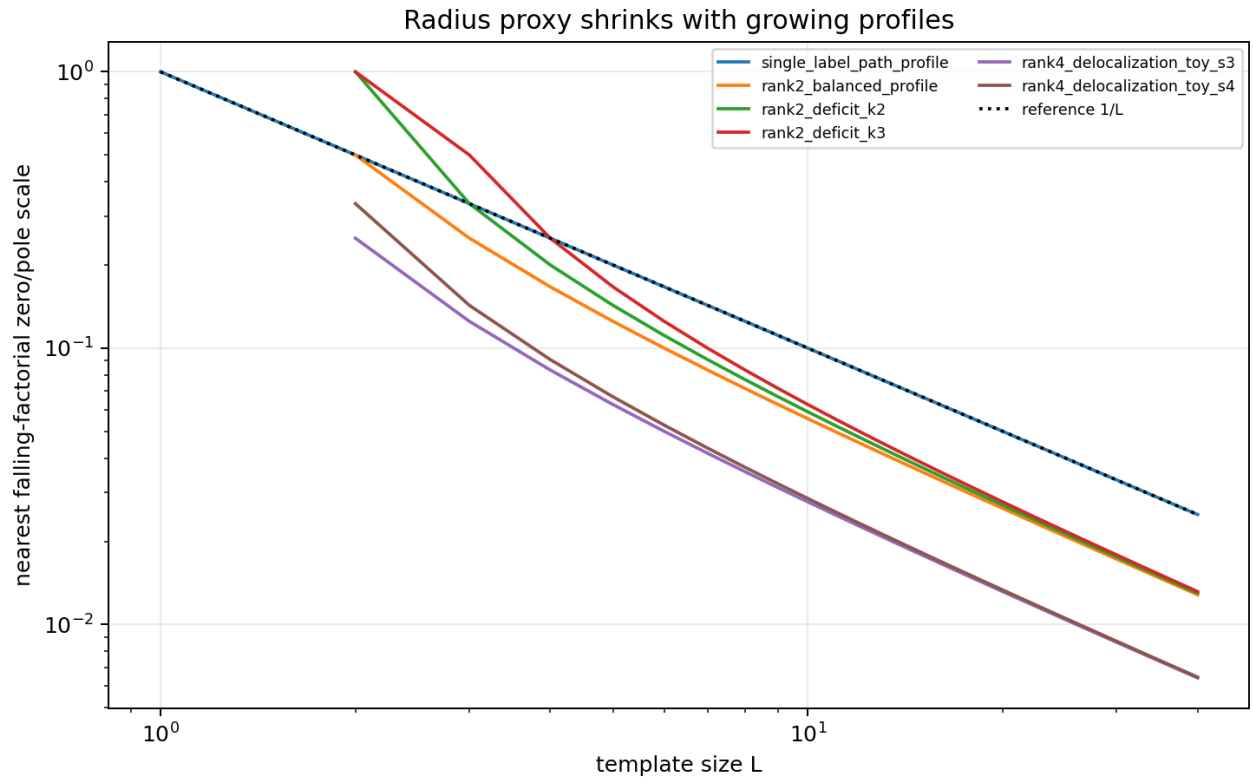


Figure 5: Nearest falling-factorial zero/pole scale versus L .

Finding 4: M5 Is Validated as a Benchmark Principle, Not as a Theorem Improvement

The cumulative M5 result has three validated layers:

1. Candidate ranking identified Markov/interpolation loss as the most tractable extension path.
2. Fixed-template expansions showed normalized conflict-free expectations are analytically tame near $x=0$.
3. Growing-template profiles showed coefficient and derivative amplification appears naturally when profile size grows.

This supports a precise research narrative: the finite-template mechanism is stable for fixed templates, but Kim-Tao-like interpolation difficulty reappears when the relevant polynomial family grows in size, support, or degree.

The report does not claim that the Kim-Tao rigidity exponent can be improved. It also does not replace imported MPvH, Nau, or MP23 inputs. Its contribution is a reproducible benchmark that separates fixed-template stability from growing-family derivative amplification.

Discussion

Cycles 13-15 converted the campaign from broad extension search to a narrow, testable mechanism.

The useful distinction is between two sources of instability:

- **Sparse-grid reconstruction instability:** high-degree fits from few reciprocal- n samples can have large coefficient and derivative norms even when the underlying fixed-template expectation is stable.
- **Growing-family analytic instability:** when template size grows, the falling-factorial factors introduce zeros and poles closer to $x=0$, and derivatives can grow rapidly.

This distinction matters because Kim-Tao's proof does not control one fixed finite template. It controls trace/pre-trace polynomial families whose degree, support, or combinatorial size grows. The M5 benchmark therefore does not solve the paper's bottleneck, but it identifies where a finite exact model first begins to resemble that bottleneck.

The most defensible next step is not more random sampling. It is a compact M5 synthesis or a transition to M6-final-synthesis, preserving the benchmark principle and its limitations.

Open Questions

1. Can the growing-template coefficient growth be bounded in a clean asymptotic form in L and expansion order r ?
2. Which growing profiles best model the actual trace-side and fourth-moment polynomial families used in Kim-Tao?
3. Can the fixed-template expansion lemma be stated and certified in a more formal system, or is the current Wolfram/Python validation sufficient for its role?
4. Does the benchmark help isolate a technical improvement path inside the imported MPvH/Nau/MP23 machinery, or does it only explain why the existing Markov loss is structurally expected?
5. Should the next cycle close M5 with a synthesis, or begin M6-final-synthesis with the benchmark as the campaign's main extension result?

References

No REFERENCES.md file was present. The report uses the following local sources and session records.

- Kim-Tao local paper files: 2603.01127.pdf, 2603.01127.txt.
- Cycle 13 sessions: d1abcc72-61a9-4c04-87c4-9d033a1fdb4, 97208720-81aa-472a-9a23-6c75eb3a4ab5, 9b4dad9-3ce9-4d3b-9dff-6b5ad699c58c.
- Cycle 14 sessions: a24e3475-402a-40e6-842f-a64f72049074, 17268f9e-adbc-44ae-90f5-0dce6fd335dd, 30702dc0-9920-45f8-8f36-b8e9d6c310c6.
- Cycle 15 sessions: bcd1ac47-e279-449f-a77d-dff1d48cddb5, 66c94dcb-f429-4d34-84f4-d9ef7d8e2f33, bb04c33d-3042-4fad-a335-eaee10bdea9f.
- M5 reports: reports/extension_candidates/m5_extension_candidate_ranking.md, reports/extension_candidates/m5_primary_candidate_statement.md, reports/extension_candidates/m5_falling_factorial_expansion_test.md, reports/extension_candidates/m5_growing_template_expansion_growth.md.
- M5 data: data/extension_candidates/m5_candidate_scores.csv, data/extension_candidates/labelled_embedding_expansion_coefficients.csv, data/extension_candidates/labelled_embedding_expansion_fit_comparison.csv, data/extension_candidates/growing_template_expansion_coefficients.csv, data/extension_candidates/growing_template_expansion_summary.csv.

Appendix: Implementation Details

Code Organization

New Cycle 13-15 scripts:

File	Lines	Purpose
scripts/score_m5_extension_candidates.py	183	Regenerates the M5 candidate score CSV and candidate matrix figure.
scripts/derive_labelled_embedding_expansions.wls	75	Exports fixed-template falling-factorial expansions through x^4 .
scripts/compare_expansions_to_cycle9.py	296	Compares exact expansions against Cycle 9 polynomial-window fits.
scripts/probe_growing_template_expansions.wls	154	Exports growing-template coefficient expansions through x^8 .
scripts/plot_growing_template_expansions.py	371	Regenerates growing-template CSVs and figures.

New Cycle 13-15 tests:

File	Lines	Purpose
tests/test_labelled_embedding_expansions.py	83	Tests fixed-template coefficients, conflict controls, truncation behavior, and missing-fit skip handling.
tests/test_growing_template_expansions.py	80	Tests growing-template hand formulas, truncation agreement, deterministic extraction, and radius monotonicity.

Data and Figures

Cycle 13 produced data/extension_candidates/m5_candidate_scores.csv with seven ranked candidates and reports/figures/m5_extension_candidate_matrix.png at 1890 x 1007.

Cycle 14 produced data/extension_candidates/labelled_embedding_expansion_coefficients.csv with seven template rows, data/extension_candidates/labelled_embedding_expansion_fit_comparison.csv with 581 lines including header, and reports/figures/m5_expansion_vs_cycle9_fits.png at 2700 x 864.

Cycle 15 produced data/extension_candidates/growing_template_expansion_coefficients.csv with 2494 lines including header, data/extension_candidates/growing_template_expansion_summary.csv with 278 lines including header, and three figures: m5_growing_template_coefficient_growth.png at 1440 x 900, m5_growing_template_derivative_growth.png at 2160 x 900, and m5_growing_template_radius_proxy.png at 1440 x 900.

Validation Results

Cycle 13 validation passed:

- python3 -m py_compile scripts/score_m5_extension_candidates.py
- python3 scripts/score_m5_extension_candidates.py
- figure readability check for m5_extension_candidate_matrix.png
- promise_check and org_check, both exit 0 with known warnings only

Cycle 14 validation passed:

- wolfram-batch -script scripts/derive_labelled_embedding_expansions.wls
- python3 -m py_compile scripts/compare_expansions_to_cycle9.py tests/test_labelled_embedding_expansions.py
- python3 scripts/compare_expansions_to_cycle9.py
- python3 tests/test_labelled_embedding_expansions.py
- figure readability check for m5_expansion_vs_cycle9_fits.png
- promise_check and org_check, both exit 0 with known warnings only

Cycle 15 validation passed:

- wolfram-batch -script scripts/probe_growing_template_expansions.wls
- python3 -m py_compile scripts/plot_growing_template_expansions.py tests/test_growing_template_expansions.py
- python3 scripts/plot_growing_template_expansions.py
- python3 tests/test_growing_template_expansions.py

- figure readability checks for all three growing-template figures
- promise_check and org_check, both exit 0 with known warnings only

After this reporter pass, MANIFEST.md was updated to include the M5 artifacts. Follow-up validation reported promise_check exit 0 with 32 ledger events and known warnings only, and org_check exit 0 with known workspace-organization warnings only.

Cumulative Snapshot

The updated manifest records:

Metric	Value
Campaign scripts	14
Campaign script lines	3,394
Campaign test files	8
Campaign test lines	576
Markdown/DOT/PNG documentation artifacts under docs/ and reports/	60
PNG figures under reports/figures/	20
Canonical CSV datasets under data/	22
Promise ledger events	32

Current milestone state:

- M1-paper-map: validated.
- M2-proof-ledger: validated narrowly for local proof reconstruction and quantitative dependency/loss accounting.
- M3-computational-probes: validated as a reproducible finite random-permutation and Schreier-operator benchmark suite.
- M4-formal-certification: validated for the labelled-template embedding expectation identity.
- M5-extension-candidates: validated through candidate ranking, fixed-template expansion testing, and growing-template expansion growth benchmark.
- M6-final-synthesis: pending.

Cross-Reference Map

Origin	Consuming artifact	Role
docs/proof_ledger/m2_loss_map.md	m5_extension_candidate_ranking.md	Identifies Markov/interpolation loss as the primary extension target.
reports/computational_probes/m3_polynomial_window_diagnostics.md	m5_primary_candidate_statement.md	Supplies low-degree stability and high-degree instability evidence.
reports/formal_certification/labelled_embedding_expectation_identity.md	m5_falling_factorial_expansion_test.md	Supplies the exact M4 falling-factorial identity.
derive_labelled_embedding_expansions.wls	labelled_embedding_expansion_coefficients.csv	Exports fixed-template coefficients.

Origin	Consuming artifact	Role
compare_expansions_to_cycle9.py	m5_expansion_vs_cycle9_fits.png	Compares exact expansions with Cycle 9 fits.
m5_falling_factorial_expansion_test.md	probe_growing_template_expansions.wls	Motivates varying template size after fixed-template stability is established.
probe_growing_template_expansions.wls	growing_template_expansion_coefficients.csv	Exports growing-profile expansion data.
plot_growing_template_expansions.py	growing-template figures	Renders coefficient, derivative, and radius-proxy evidence.